

Speakers

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for the insurance companies to gauge what can break down in an accident and how much they will have to pay. Accordingly, they plan the premium of the insurance. But for EVs, as the concept is still in its early adoption stage in many countries including India, it is relatively not that easy for insurance companies to assess premium.

However, a lot of EV companies in the world, including Tesla, collect a lot of data around ownership experience. This includes mileage, speed, every location, and time at which the car gets charged, and more. In some cases, there are also video clips captured from the car's external cameras, which could include the car's surroundings and driving conditions. This data, as Ranjita explains, can be a good encouragement point for the insurance companies for head start.

What Can EV Makers Do to Encourage Adoption

1. Offer plans such as battery swapping. It may make the end consumer feel risk-free while investing in an EV.
2. Design should include safety from battery theft.
3. Collect as much data as is relevant for financial institutions and insurance companies to show them return on their investments.
4. Four-wheelers offer a good opportunity, just like three- and two-wheelers.
5. India is a price-sensitive market, so products should be developed keeping in mind a specific target audience.
6. Build supply chains for raw materials. Small changes in raw material cost can create havoc in pricing strategy.
7. Work on charging infrastructure as a part of the product development. If consumers can not charge their EVs, why would they invest in the same!
8. Explore OPEX as well as CAPEX models. Different consumers might want to experience EVs through different pricing models.

"Such data is something that still does not exist in EVs. We as OEMs can go on and on and say that there are lesser moving parts in EVs than in ICE cars, but the insurance and financing bodies need substantial amounts of data to ascertain this fact. The statistical models around EVs that would make insuring bodies' traditional ways work are not just there," explains Ranjita.

She adds, "We solved this problem by following an innovative idea that the Orxa Energies team came up with. Over the lockdown, we built a system that can track a battery pack and can be installed onto fleet operators. The data collected can be made accessible to our financing partner."

Ranjita further explains that Orxa's financing partner buys battery packs from them and then leases the same to fleet operators on per day, per week, per month, and so on basis. Through the data being fed by the system, the partner can know whether the fleet partner is using the battery packs cautiously or not.

"Talking about the levels of data, we have hundreds and hundreds of data points that can be made available to us and the partner. Now the question that needs to be asked is, which part of the data collected by this battery management system is relevant for whom. **You can give financing people the data that can help them make decisions on financing EVs and similarly data to insurance companies for insuring EVs,**" says Ranjita.

Financial institutions finance products following one condition, which is what kind of returns they

are getting. In Delhi-NCR, a lot of financial institutions, as per Verma, had actively participated in the first round of e-rickshaw or three-wheeler sales. However, these financial institutions ran into two classical problems during the process. The first one was the customer's inability to pay and unwillingness to pay. A lot of these were related to battery packs in one way or the other.

"There were customers who were not willing to pay and then there were customers who were not able to pay. Why did that happen? Because of the theft of the battery packs from these e-rickshaws. Some statistics suggested that almost 28 per cent of lead-acid battery packs were getting stolen. Now no insurance company will underwrite this heavy a burden," explains Verma.

He adds, "The second problem was the battery packs starting to underperform six to eight months down the line. Now the guy who had invested in buying one had to invest again in battery packs. So, a lot of such guys just defaulted on the payments."

The solution to these problems, as per Verma, is a push towards lithium-ion battery packs instead of lead-acid battery packs. The former is much more efficient than the latter. The former also have a longer life than the latter. Using lithium-ion battery packs can also help in controlling theft.

"Lithium-ion battery packs are not an easy problem to solve, but arguably it is a much easier problem to solve than the ones associated with lead-acid battery packs," notes Verma. **EFY**